

Evidence — Study Guide

Sessions 1–2 · one guide per type of evidence

Team-Based-Learning course · core concepts + diagrams + glossaries + iRAT-prep practice

INFERENCE

QUALITY

RELEVANCE

S1

Welcome: Context & the Grand Challenge

framework

S2

Descriptive Quantitative Evidence

descriptive · quantitative

Welcome to Evidence: Context & the Grand Challenge

Each form of evidence has a *unique affordance* — it answers a different question and licenses a different inference. Learning to judge **quality** and **relevance** is how we keep those inferences sound.

evidence framework

inference · quality · relevance

ladder of inference

paradigms

01 The core idea — the Evidence Framework

The big idea for the whole course, and this session: **each form of evidence has a unique affordance**. Each type answers a different question and lets us make a different *type* of inference. The central skill is deciding whether you are using the *right type of evidence* for the inference you want to make.

An **affordance** is what a given type of evidence *lets you do* — the kind of question it can answer and the kind of conclusion it can support. Descriptive evidence can tell you *what is happening*; it cannot, on its own, tell you *what caused it*. Match the evidence to the claim, or the inference breaks.

The four types of evidence

You'll draw on all four to build your final recommendation to São Paulo. A single study can contain several types at once (e.g., Calarco's classroom study is mostly descriptive-qualitative but also reports descriptive-quantitative counts of help requests).

TYPE	QUESTION IT ANSWERS	TYPICAL METHODS
Descriptive	What is happening? (e.g. what is happening with income inequality in Brazil?)	Quantitative (numbers describing a population) or qualitative (words describing a local setting)
Causal	Is a policy, program, or practice causing an impact?	Experiments and quasi-experiments
Synthetic	What do we already know about this topic?	Synthesizes / combines prior research
Process	Not what is happening, but <i>how</i> and <i>why</i> ?	Studies of mechanisms and implementation

The three throughlines of the whole course

Every session returns to these three ideas. They are the lens you use on every reading.

Inference THE GOAL

A conclusion drawn on the basis of reasoning and evidence. The course is one long exercise in making **sound** inferences — and noticing when a claim outruns its evidence.

Ask: what conclusion does this evidence actually license?

Quality HOW STRONG

How well the claims are supported by the *type* of evidence used. Each type is judged on **its own terms** — but two questions apply to all: how well-supported are the claims, and what are the limitations?

Ask: what claims can we *not* make from this?

Relevance

APPLIES TO ME?

Whether high-quality findings actually apply to *your* context. Evidence gathered "there and then" must be reasoned into "here and now" for São Paulo's challenge.

Ask: how similar are the setting & sample — and would the differences matter?

EVIDENCE
a specific type



AFFORDANCE
the question it can answer



INFERENCE
a matching conclusion

Right type of evidence → the inference holds. Wrong type → the inference breaks.

02 The Grand Challenge

The course is organized around one real problem — the **Grand Challenge** — and it is also your final, culminating assignment. In the spirit of a "no surprises" policy, you see the final before class even begins.

The Grand Challenge: How can we improve academic achievement in the foundational subjects of **math, reading, and science** for students in **years 6–9** in the **São Paulo** school system? Underneath it sits the field's larger question: *how do we improve learning and advance equity in pursuit of a more just society?*

The Brazilian context (from the OECD reading)

Brazil's education system has made great progress over the last couple of decades — the vast majority of children are now enrolled in basic education. But there are **great disparities in access and achievement along class lines**, and these **deepened during the COVID-19 pandemic**. Much of the reading is *descriptive quantitative* evidence about income inequality.

You will use evidence in four ways

Your final recommendation to the São Paulo Secretary of Education has four moves — and each pulls on the evidence framework:

1

Frame the issue — highlight the dimension of the problem people should focus on.

FRAMING

Descriptive evidence helps you decide *what* the problem really is before proposing fixes.

2

Discuss considerations for expanding a curricular innovation to São Paulo.

RELEVANCE

Was the innovation shown to work elsewhere? Would it transfer "here and now"?

3

Recommend a form of teacher coaching to support that expansion.

CAUSAL + PROCESS

What coaching causes improvement, and how/why does it work?

4

Recommend additional evidence São Paulo should collect to keep improving.

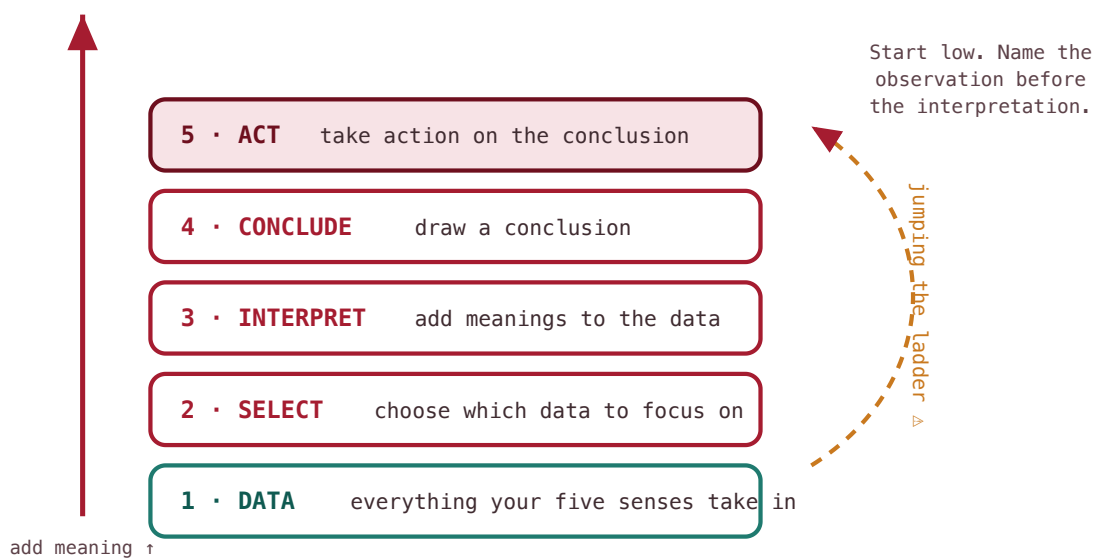
AFFORDANCES

Which type of evidence fills the gaps that current evidence can't?

03

The Ladder of Inference & confirmation bias

One way to picture inference is as a **ladder**. It breaks the move from raw data to action into steps — and shows where we tend to slip. The farther you climb from the data, the more careful you must be that your conclusion is actually warranted.



Data → select → interpret → conclude → act. The higher the rung, the more the conclusion depends on the type of evidence being adequate.

The data at the bottom aren't only research findings — they also include your **professional expertise, personal relationships, and knowledge of local context**, and can be quantitative (numbers) or qualitative (words, stories). Our brains are wired to make meaning out of data — that's how we make sense of the world — but it makes it dangerously easy to **jump up the ladder**, adding interpretation and conclusions without supporting data.

Worked example — the OECD report (p. 6)

The authors model the ladder in a single passage:

CONCLUSION FIRST → "Brazil is marked by deep-rooted inequalities by race and class." (*How do we know we can trust this?*)

SELECTED DATA → In 2018, Black and mixed Brazilians were more than half the population but over 75% of the poorest income decile.

INTERPRETATION → Not just overrepresentation — little indication the race-based gap will narrow (Figure 1.3).

CONCLUSION / ACTION → Inequality is deep-rooted and intersectional — grounds for specific education interventions in your Grand Challenge.

Nuance & watch-outs

- ⚠ **Don't jump the ladder.** It's easy to add interpretation and draw conclusions without the supporting data. Start on the lowest rung: *name your observation before* stating your interpretation, conclusion, or action.
- ⚠ **Confirmation bias runs in one direction.** We readily spot where an author "jumped the ladder" when we *disagree* with the conclusion — but wave through unjustified inferences when we *agree*. Staying low on the ladder forces you to clarify the data you're actually relying on, and works against this very human error.
- ⚠ **"There and then" ≠ "here and now."** High quality is not enough — evidence from another setting still has to be reasoned into São Paulo's context. Non-São-Paulo data isn't irrelevant; it just takes more work to see how it applies.
- ⚠ **No evidence type is "best."** Affordances differ. Judge each type on its own terms — and check you're using the right type for the inference you want.

Paradigms — the worldview behind the evidence

Who decides what counts as good evidence? That depends on the researcher's **paradigm** — the worldview that structures their approach to knowledge, their research goals, and how they judge quality. Two researchers in different paradigms can look at the *same* data and genuinely see different things.

Think of the **duck-rabbit** picture: the same image reads as a duck or a rabbit depending on the lens you bring. Likewise, a paradigm sets the conventions for what data to collect, how to analyze it, what counts as convincing, and which inferences you can draw. Know a researcher's paradigm *before* you judge their work — so you apply the conventions *they* were working under.

This course adapts the classification of **Lincoln, Lynham & Guba** into five common paradigms. You don't need to memorize the details — the point is to feel the *range*. Two distinctions separate them: their views on **whether/how objectivity is possible**, and on **the researcher's influence on the research**.

Positivism

STABLE TRUTH • DISCOVER IT

A single objective reality exists and can be uncovered through careful measurement. Patterns are independent of the researcher's values. Common in the hard sciences and experimental psychology.

Post-positivism

TRUTH, BUT ONLY PARTIAL

There is an objective truth, but we capture only facets of it — with error — by **triangulating** multiple data sources. Most course readings, and many quantitative studies, live here.

Interpretivism

HOW DO PEOPLE INTERPRET IT?

Cares about the worldviews of those studied — and how the researcher's own values shape their reading of them. Asks not "what is happening?" but "how do people interpret what's happening?" Common in qualitative evidence.

Participatory

KNOWLEDGE CO-CONSTRUCTED

Researchers and participants make sense of the world *together*. Questions whether an "outsider" should try to interpret others' worldviews at all — knowledge is a shared project.

Critical

WHO HAS POWER & VOICE?

Unearths the social, political, and economic forces shaping who speaks and who is heard. Overlaps with participatory, but adds explicit critique of power structures — and often argues for redistributing power.

Key points about paradigms

- △ **No paradigm is more objective, valid, or biased than another.** They differ on whether/how objectivity is possible — and different biases surface in different ones.
- △ **Paradigms are loose frameworks, not fixed boxes.** A researcher can hold several at once, and multiple can appear in one piece of research.
- △ **We need multiple paradigms.** Fully addressing educational challenges takes evidence from many perspectives — a strength of education research.
- △ **Where does Evidence sit?** The course takes a **post-positivist** stance — *not* because it's "best," but because it's dominant in education policy and practice, so it prepares you to work within it beyond HGSE. (That dominance is historical — e.g., Cold-War-era economic thinking — and has meant qualitative, participatory, and critical traditions often held less power in policy settings.)

✦ Readings for this session



OECD, *Education in Brazil* (2021) — Chapter 1. Introduces the current state of Brazilian education: strong enrollment progress, but deep disparities by race and class, worsened by COVID-19. Mostly *descriptive quantitative* evidence — and a live example of the ladder of inference.

Pages 16–33 only · skip the "emerging issues" section · via course Library Reserves



iRAT prep — test yourself

Answer each before revealing. Two are TBL-style multiple choice; the rest are short-answer / application. Two discussion prompts follow.

Q1 · MC

The "unique affordance" of a type of evidence refers to...

REVEAL ►

Q2 · MC

A researcher assumes there is an objective truth but that we can only capture facets of it, with error, by triangulating multiple data sources. Which paradigm?

REVEAL ►

Q3 · SHORT ANSWER

List the rungs of the ladder of inference in order, and state the one rule the video gives for climbing it well.

REVEAL ►

Q4 · APPLICATION

You strongly agree with a study's conclusion, so you accept it quickly without scrutinizing its evidence. Name the reasoning error and the habit that counters it.

REVEAL ►

Q5 · APPLICATION

A well-designed U.S. study shows a math program raised achievement. Before recommending it for São Paulo, which throughline concept must you apply, and what two questions does it ask?

REVEAL ►

Discussion prompts

1. Pick one figure or claim from the OECD reading. Walk it up the ladder of inference: what is the raw *data*, what did the authors *select*, how did they *interpret* it, and what *conclusion / action* follows? Where, if anywhere, might they be climbing too fast?
2. The course adopts a post-positivist stance while insisting all paradigms are equally valid. Is there tension there? For São Paulo's Grand Challenge, when might interpretivist, participatory, or critical evidence reveal something post-positivist evidence would miss?

§ Glossary — quiz yourself first

Try to define each term before you reveal it. These are the Session 1 vocabulary plus key terms from the videos.

inference

REVEAL ►

affordance

REVEAL ►

quality

REVEAL ►

relevance

REVEAL ►

ladder of inference

REVEAL ►

confirmation bias

REVEAL ►

the four evidence types

REVEAL ►

paradigm

REVEAL ►

post-positivism

REVEAL ►

triangulation

REVEAL ►

★ Remember this

- ◆ Each form of evidence has a unique affordance — it answers a different question and licenses a different inference. Use the right type for the claim you want to make.
- ◆ Four types: descriptive (what?), causal (is X causing it?), synthetic (what do we know?), process (how & why?). A single study can mix them.
- ◆ Three throughlines run the whole course: inference (draw sound conclusions), quality (how well evidence supports its claims, judged on its own terms), and

relevance ("there and then" → "here and now").

- ◆ **Climb the ladder carefully.** Data → select → interpret → conclude → act. Name the observation before the interpretation, and you'll dodge confirmation bias.
- ◆ **Paradigms are lenses, not rankings.** No paradigm is more valid or biased; the course works in post-positivism because it dominates policy — while treating all paradigms as equally valid.
- ◆ **The Grand Challenge:** improve math, reading & science achievement for São Paulo students in years 6–9 — your culminating assignment, built from all four evidence types.

Descriptive Quantitative Evidence

The first form of evidence: describing *what is* — in numbers. Big idea of the day: **measure the few to learn about the many.**

population ↔ sample

random sampling

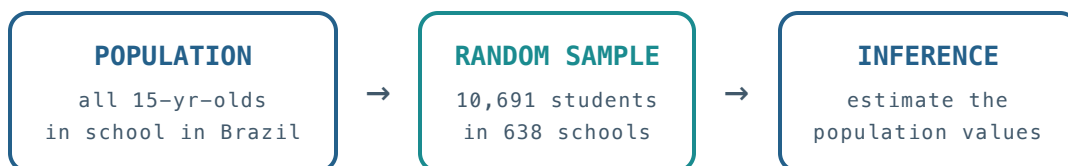
typical values & variability

plausible values

01 The core idea

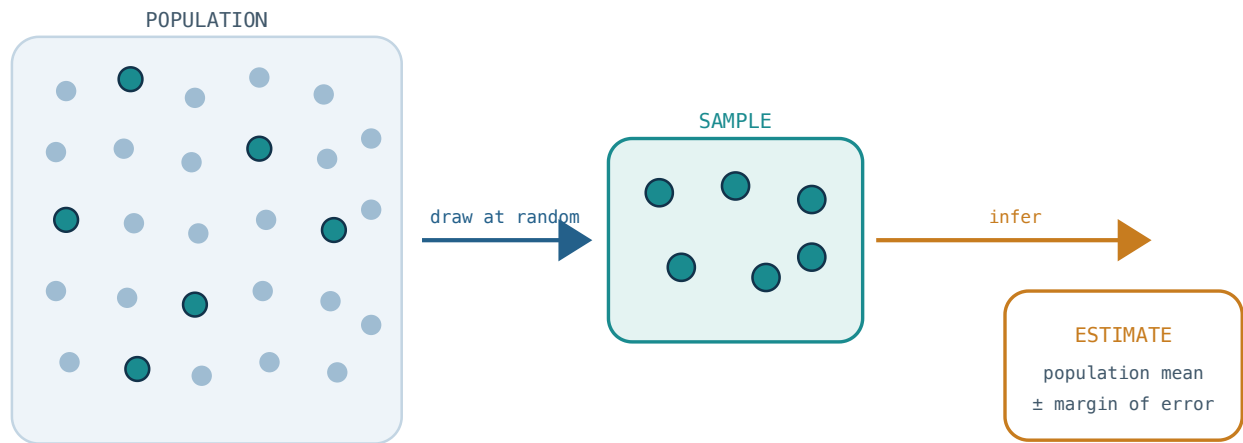
Descriptive evidence answers one question: "**What is happening?**" Descriptive *quantitative* evidence answers it with numbers (the qualitative flavor answers it with words). It does **not** explain causes — only describes how things are.

We almost never measure everyone. Instead we name a **target population** — the group we want to describe — draw a smaller **sample** from it, calculate statistics on the sample, and use them to make an **inference** back to the population. The more representative the sample, the more trustworthy the inference. That is the whole logic of the day.



Sampling logic: we measure the few (the sample) to learn about the many (the population).

In the reading, the OECD uses two data sources on Brazil: **PISA** (a global assessment of reading, math, and science for 15-year-olds) and **SAEB** (Brazil's national assessment). The headline descriptive findings: Brazilian students score far below the OECD average in all three subjects, roughly on par with Latin American neighbors, but below economically similar countries like Russia and China.



The engine of descriptive quantitative evidence: population → random sample → statistic
→ inference (with a margin of error).

Three quality questions for this kind of evidence

Every form of evidence has its own affordances, so we judge each on its own terms. For descriptive quantitative evidence, we ask three questions — they organize the rest of these notes:

Q1

How well does the sampling approach support inferences about the target population?

SECTION 02

Random sampling from the whole population is what licenses the inference.

Q2

How well do the authors describe the typical values and their variability in the sample data?

SECTION 03

Report central tendency (mean/median) *and* spread (range/SD) — never one alone.

Q3

How well do the authors describe the range of plausible values for the population characteristic (and statistical significance)?

SECTION 04

A sample estimate always comes with uncertainty — report it.

02 Sampling — why random selection earns the inference

People often distrust small samples: "How can a poll of a few thousand speak for me if I wasn't asked?" The answer is **random sampling**.

If we draw a **sufficiently large sample at random from the whole population**, the proportion in the sample tends to land close to (though never exactly on) the proportion in the population. If 55% of voters favor a candidate, a large random sample will usually come in near 55%. Every sample differs a little from its population by chance — even when researchers do everything right — but random selection keeps that difference small and, crucially, *unbiased*.

Two conditions have to hold together:

CONDITION	WHAT IT MEANS	WHY IT MATTERS
Selected at random	People enter the sample by pure chance — like a coin flip — not by their own choice or the researcher's convenience.	Removes systematic reasons the sample would differ from the population.
From the whole population	Drawn from the entire group we want to describe — not a narrower slice of it.	Lets us generalize to <i>that</i> population, not some other one.

 Where it breaks: selection bias

When sampling is **not** random or **not** from the whole population, you get a **sample of convenience** and there's no reason to think it represents anyone in particular. Classic offenders:

-  People who live near a university and volunteer to participate.
-  Psych students earning extra credit.
-  Whoever finds a website and answers its questions.
-  Poll respondents who answer the phone — missing everyone who screens calls.

Applied to the reading: who does the PISA sample actually represent?

The report (p. 71) describes its sample: **10,691 fifteen-year-olds across 638 schools**. How was it drawn? PISA can't test every student in every country (too expensive), so each country selects a **random sample of schools**, then a **random sample of students within those schools**.

Because selection is random, the sample is a good stand-in for the population of *enrolled* 15-year-olds.

But here's the catch the video flags: about **one-third of Brazilian 15-year-olds are not enrolled in school**, and the sample only includes enrolled youth. So we can infer about **15-year-olds enrolled in school in Brazil** — *not* about all Brazilian 15-year-olds.

Random selection tells you the sample is fair; it's the *frame* (the whole population you sampled from) that tells you who you're allowed to talk about.

03 Describing data — typical values *and* variability

Once you have sample data, two questions describe it well: **What's typical?** and **How much do the values vary around that?** A good description answers both. Report only one and you can seriously mislead.

Typical values CENTRAL TENDENCY

Mean — the average. Add up all values, divide by the number of observations.

Median — the midpoint. Order values low to high; take the middle one (the 50th percentile).

Read it as: if you see mean or median cited, that's the author telling you what a typical case looks like.

Variability SPREAD

Range — the distance from the lowest to the highest value.

Standard deviation — roughly, the typical distance of a data point from the mean; how tightly points cluster around it.

Read it as: if you see range or SD, that's the author telling you how much the data spreads out around the typical value.

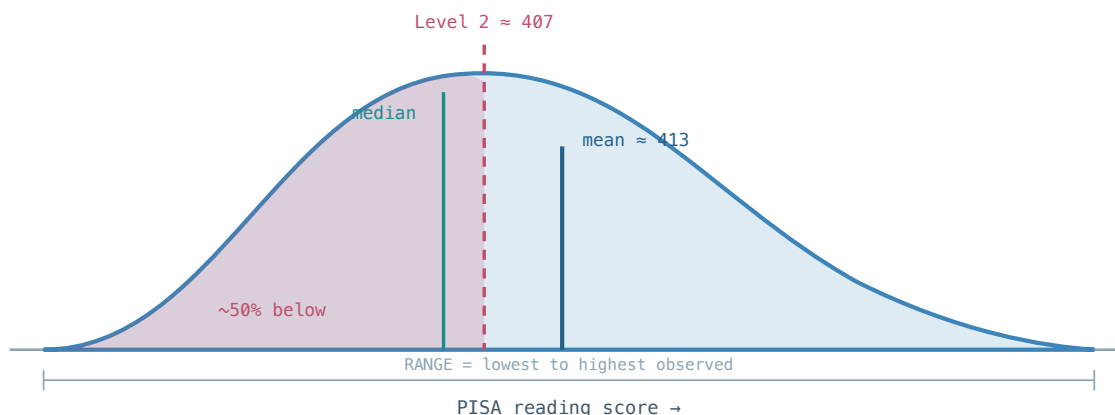
↳ You don't need to *calculate* any of these for EVI101. You need to *recognize* which job each one does: mean/median describe the center; range/SD describe the spread.

Why the average alone can fool you — Brazil's reading scores

Brazil's **average PISA reading score is 413** (Figure 3.2, p. 79). The report also notes that scoring **407 or above** (PISA Level 2) signals likely future success in the workforce. Tempting conclusion: "413 > 407, so Brazil is doing fine." That's wrong — because it ignores the spread.

Figure 3.3 shows the **full range** of how students scored. Once you look, **50% of Brazilian students score below Level 2** — below the threshold for likely success. The average sits just

over the line, but half the distribution falls under it. This is the whole point of reporting variability, not just the center.



The mean can clear a threshold while half the students fall below it. Center and spread tell different stories — read both. (In a right-skewed distribution the mean is pulled toward the long tail, sitting above the median.)

04 From sample to population — plausible values & significance

A sample statistic is a best guess, not the truth. Because the sample differs from the population by chance, the average we compute won't exactly equal the population average. Honest evidence reports the **uncertainty** around the estimate.

Range of plausible values (the margin of error)

We could know Brazil's exact average score only by testing *every* 15-year-old. Instead we use the sample to make a best estimate and attach a **margin of error** — a **range of plausible values** for the population average. PISA figures draw this as a **95% confidence interval**: a vertical bar around each point estimate.

From the 2022 PISA country note (OECD, shown in the video): for the top socioeconomic quintile in Brazil, the sample mean in math is just above 450, with a margin of about ± 5 points (roughly 445–460). That's a *useful* interval. If it had run 400–500, it would pin down almost nothing. Notice too: the **bottom quintile has a narrower interval** because its sample is larger — **bigger samples give more precise estimates**.

⚠ Don't confuse two different "ranges"

- ⚠ **Range of the data** — the spread of individual data points in the sample (Section 03). Describes how varied people are.
- ⚠ **Range of plausible values for the average** — the uncertainty in our *estimate of the population mean* that comes from using a sample. Describes how sure we are about the average.

Statistical significance

Statistical significance asks: is a difference we see in the sample likely real in the population, or could it just be random chance? One quick visual test: **compare confidence intervals**. In the 2022 figure, the interval for top-quintile students (bottom end ≈ 445) doesn't overlap the interval for bottom-quintile students (top end ≈ 350). Non-overlapping intervals signal a **statistically significant difference** — the SES gap is very unlikely to be a fluke.

The reading uses the same logic across time. In Figure 3.5, Brazil's yearly reading averages are compared to 2018 (score 413). Values flagged blue-with-asterisk are statistically significantly different from 2018: the 2000 and 2006 scores are significantly *lower*, while 2003, 2009, 2012, and 2015 are *not* significantly different. The inference: Brazilian reading improved over earlier trends by around 2009, then flattened. Effectively the authors put a 95% confidence interval around the *difference* between 413 (2018) and 396 (2000) and ask whether that 17-point gap is a chance finding.

Direction & magnitude

When you report a difference, two things describe it: **direction** (which way — higher or lower, up or down) and **magnitude** (how big). 2000 vs 2018 is a difference in *direction* (2000 was lower) and *magnitude* (about 17 points). Significance tells you the difference is probably real; direction and magnitude tell you what it actually amounts to.

⚠ Nuance & watch-outs

- ⚠ **A biased sample can't be rescued by size**. If the sample isn't random or isn't from the whole population, a bigger sample just gives you a more precise wrong answer.

- ⚠ **Mean vs median under skew.** A few extreme values drag the mean toward the tail; the median barely moves. With skewed data, the median is often the more honest "typical."
- ⚠ **"Not significant" $\neq$ "no difference."** It means any observed difference isn't large enough to rule out chance — not that the true difference is zero.
- ⚠ **Significant $\neq$ important.** A difference can be statistically significant yet tiny in magnitude and practically meaningless. Statistical significance $\neq$ practical significance.
- ⚠ **Report a range, not just a point.** A bare estimate hides its uncertainty. Always look for (or provide) the range of plausible values.
- ⚠ **Describe $\neq$ explain.** Descriptive evidence can't establish causation. Lower-income students scoring lower does *not* show income caused it — that needs causal evidence. Stay low on the ladder of inference: infer only from what the data actually measured.

05 Today's application — São Paulo's Grand Challenge

The in-class task: São Paulo leaders want to frame a Grand Challenge — *why focus on academic achievement in foundational subjects for grades 6–9?* **Which figure should they use to make the case?** There is **no single correct answer**; the point is to *use the language of evidence* to argue for one choice over the other.

OPTION A

Figure 3.5

PISA performance in reading, math, and science **over time**, benchmarked against the OECD and Latin America. Frames the challenge as a *level-and-trend* problem: how far below peers Brazil sits and how little it has moved.

OPTION B

Figure 3.10

PISA reading performance broken out by students' **socio-economic background**. Frames the challenge as an *equity* problem: outcomes diverge sharply across SES within Brazil.

How to argue it in the language of evidence: Whichever you pick, ground it in Session 2 vocabulary. *Figure 3.5* leans on **typical values and trends over time** and on **statistical significance** (which year-to-year changes are real vs chance) — good for a "we're falling behind and not catching up" story. *Figure 3.10* foregrounds **variability across subgroups** and lets you point to **non-overlapping plausible-value ranges** between rich and poor

students — good for an "our problem is inequality, not just level" story. A strong answer names the population each figure describes, what's typical, how much it varies, and what inference the framing supports — then argues why that framing best motivates the grades 6–9 focus.

06 Glossary — quiz yourself first

Try to define each term before you reveal it. These are the Session 2 evidence-vocabulary words.

random sampling

REVEAL ►

sampling logic

REVEAL ►

mean

REVEAL ►

median

REVEAL ►

range

REVEAL ►

standard deviation

REVEAL ►

selection bias

REVEAL ►

range of plausible values

REVEAL ►

statistical significance

REVEAL ►

direction

REVEAL ►

magnitude

REVEAL ►

Answer before revealing. Two are written like TBL readiness-check multiple-choice; the rest are short-answer/application. Then chew on the discussion prompts.

MC 1. A wellness app surveys the users who opt into its "share my data" feature and reports the average sleep of "adults." The biggest threat to inferring about all adults is —

[ANSWER ►](#)

MC 2. A district's household incomes include a few very wealthy families. Which single number best describes the *typical* household?

[ANSWER ►](#)

SA 3. The PISA sample was randomly drawn — so why can't we use it to describe *all* Brazilian 15-year-olds?

[ANSWER ►](#)

SA 4. Brazil's average reading score (413) sits above the Level-2 threshold (407), yet the report warns against optimism. Why?

[ANSWER ►](#)

SA 5. Two subgroups' 95% confidence intervals don't overlap. What can you infer, and what can you *not* infer?

[ANSWER ►](#)

DISCUSSION 1

The report shows lower-income Brazilian students score lower on PISA. A colleague says, "So poverty lowers test scores." Using the language of evidence, explain exactly what this descriptive data can and cannot support — and what kind of evidence would be needed to make the causal claim.

DISCUSSION 2

Imagine a finding that is statistically significant but only 2 points on a 600-point scale. How would you communicate this to a policymaker so they grasp both the significance and the magnitude without over-reacting?

For this session

- **Required:** OECD, *Education in Brazil* (2021), **Chapter 3, pages 70–92 only**. Access via the course Library Reserves page. Key figures: 3.2 (average scores), 3.3 (full range of reading scores), 3.5 (scores over time), 3.10 (scores by SES).
- **Also referenced:** **Figure 5** from the OECD PISA 2022 factsheet / country note on Brazil (shown in the session video to illustrate confidence intervals and the range of plausible values).

★ Remember this

- ◆ **Descriptive quantitative evidence describes "what is" in numbers** — it can't explain causes.
- ◆ **Measure the few to learn about the many:** a random sample from the whole population lets you infer population values.
- ◆ **Selection bias breaks inference** — and no amount of sample size repairs a non-random or wrong-population sample.
- ◆ **Describe both center and spread:** mean/median for typical values, range/SD for variability. The average alone can mislead.
- ◆ **Report a range of plausible values**, not just a point estimate — bigger samples give narrower confidence intervals.
- ◆ **Statistical significance $\neq$ practical importance**; always pair it with direction and magnitude.